INCH-POUND
MIL-I-46559B
31 January 1990
SUPERSEDING
MIL-I-46559A(MU)
30 July 1969 and
Amendment 1
16 July 1973

MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M3A2, METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M3A2 Cartridge Actuated Initiator Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specification, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Military Specifications and Standards, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CAGE Code 19200 and 19203)

8595822	Initiator, Cartridge Actuated, Metal Parts Assembly
8796541	Box, Fiberboard (Inner) Packing, Ammunition, for Initiator, Cartridge Actuated, M3A2 or M5A2
8796542	Box, Packing, Ammunition, for Initiator, Cartridge Actuated, M3A2 or M5A2
9216934	Box, Fiberboard (Outer) Alternative Packing, Ammunition, for Initiator, Cartridge Actuated, M3A2 or M5A2

U.S. AIR FORCE

TPO 00-932-5031	Initiator, Cartridge Actuated, M3A2

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5000.)

2.2 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **Materials.** All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2 **First article.** When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 **Raw material.** The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.4 **Physical properties.** Physical properties required by the drawing shall be uniform throughout the part within specificed tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 **Hydrostatic pressure.** The subassembly consisting of a cap, chamber, and plug simulating the firing pin housing, without firing pin hole and with an O-ring, shall withstand a constant internal pressure of 5500 to 6500 pounds per square inch (psi) for 15 seconds, minimum (min), without application of any external load other than that applied by the liquid pressure and shall show no evidence of leakage or mechanical failure (see 4.6.2).

3.6 **Firing pin release.** The firing pin of the assembled initiator, less safety pin, cartridge, cartridge retainer, and chamber, shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin ¾ inch, maximum (max).

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3.7 **Initiator pin security.** The initiator pin of the assembled initiator, less cartridge, cartridge retainer, and chamber, shall not withdraw beyond the safety pin when subjected to an axial load of 200 to 220 pounds. Application of the load shall not cause permanent deformation of the safety pin.

3.8 **Metal parts assembly.** Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.9 **Protective finish.** Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.10 **Residual magnetism.** The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using the procedures in 4.6.6.

3.11 **Workmanship.** The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.11.1 **Metal defects.** All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.11.2 **Burr.** No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.11.3 **Foreign matter.** No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 **Quality assurance terms and definitions.** Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 **Inspection.** Inspection shall be in accordance with MIL-A-2550.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 68 to 77 °F
b. Altitude: Sea level
c. Humidity: 50 percent relative, max
d. Atmospheric pressure: 14.7 psia.

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish, and 50 shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until such time as an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 initiators, max, provided the homogeneity requirements of inspection lots have been retained.

4.5.2 Examination. One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability.

4.5.2.1 Retainer, Cartridge, Drawing 8593013, covering a detail of Drawing 8595822.

Characteristic	Defect	Method of Inspection
Critical		
1	Metering slots in base missing	Visual

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Characteristic	Defect	Method of Inspection
Major		
101	Pitch diameter of thread, max	Gage
102	Minor diameter of thread, max	Gage
103	Concentricity of cavity with pitch diameter of thread	Gage
104	Perpendicularity of pitch diameter of thread with base	Gage
105	Symmetry of metering slots in base with body	Gage
106	Diameter of cavity, max	Gage
107	Depth of cavity	Gage
108	Depth of bottom of counterbore	Gage
109	Depth of metering slots in base	Gage
110	Total length	Gage
111	Length of effective thread, min	Gage
112	Width of metering slot, on base	Gage
113	Internal radius at mouth of cavity missing or incorrect	Gage
114	Drill point on base, max	Gage
115	Number of elongated slots incorrect	Visual
116	Thread damaged or not full	Visual
117	Metal defective	Visual
118	Protective finish inadequate or defective	Visual
119	Burr	Visual
120	Foreign matter present	Visual
Minor		
201	Pitch diameter of thread, min	Gage
202	Diameter of body	Gage
203	Diameter of cavity, min	Gage
204	Diameter of head	Gage
205	Length of elongated slots	Gage
206	Length of head	Gage
207	Width of elongated slots	Gage
208	Location of elongated slots	Gage
209	Angle between head and body	Gage
210	Location of metering slots	Visual
211	Profile of elongated slots	Visual
212	Marking missing, incorrect, or illegible	Visual
213	Radius missing or incorrect, except as otherwise classified	Visual
214	Surface finish improper except as otherwise classified	Visual

4.5.2.2 Chamber, Drawing 8593014 covering a detail of Drawing 8595822.

Critical		
None defined		
Major		
101	Pitch diameter of large thread, min	Gage
102	Major diameter of large thread, min	Gage
103	Pitch diameter of small thread, min	Gage

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Characteristic	Defect	Method of Inspection
104	Major diameter of small thread, min	Gage
105	Concentricity of counterbore with pitch diameter of large thread	Gage
106	Concentricity of datum diameter with small thread	Gage
107	Diameter of counterbore	Gage
108	Depth of cavity	Gage
109	Thickness of wall, min	Gage
110	Location of datum diameter	Gage
111	Profile of gas port	Gage
112	Thread damaged or not full	Visual
113	Chamfer at base of cavity missing or incorrect	Visual
114	Metal defective	Visual
115	Protective finish inadequate or defective	Visual
116	Surface finish at end of gas port improper	Visual
117	Burr	Visual
118	Foreign matter present	Visual
Minor		
201	Pitch diameter of large thread, max	Gage
202	Pitch diameter of small thread, max	Gage
203	Diameter of cavity	Gage
204	Diameter of hole	Gage
205	Diameter of small thread pilot	Gage
206	Diameter of large thread pilot	Gage
207	Depth of counterbore	Gage
208	Total length	Gage
209	Length of large thread pilot	Gage
210	Length of end of large thread	Gage
211	Length of gas port	Gage
212	Thickness of wall, max	Gage
213	Location of small thread	Gage
214	Marking missing, incorrect, or illegible	Visual
215	Radius or chamfer missing or incorrect	Visual
216	Surface finish improper, except as otherwise classified	Visual

4.5.2.3 Pin, Initiator, Drawing 8595403, covering a detail of Drawing 8595822.

Critical		
None defined		
Major		
101	Concentricity of groove in head with body	Gage
102	Symmetry of head with body	Gage
103	Diameter of body	Gage
104	Diameter of groove in body, min	Gage
105	Diameter of groove in head, min	Gage
106	Length to shoulder at head, min	Gage
107	Location of groove in head	Gage

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Characteristic	Defect	Method of Inspection
108	Location of groove in body	Gage
109	Profile of groove in body	Gage
110	Profile of groove in head	Gage
111	Metal defective	Visual
112	Protective finish inadequate or defective	Visual
113	Surface finish on body and body groove improper	Visual
114	Burr	Visual
115	Foreign matter present	Visual
Minor		
201	Diameter of groove in head, max	Gage
202	Diameter of pull hole	Gage
203	Diameter of undercut, min	Gage
204	Length to shoulder at head, max	Ggae
205	Width of head	Gage
206	Width of shoulder at body groove	Gage
207	Width of groove in head	Gage
208	Width of undercut, max, if applicable	Gage
209	Total length	Gage
210	Thickness of head	Gage
211	Location of pull hole	Gage
212	Angle at base of body	Gage
213	Profile of head	Gage
214	Marking missing, incorrect, or illegible	Visual
215	Radius or chamfer missing or incorrect	Visual
216	Surface finish improper, except as otherwise classified	Visual

4.5.2.4 Cap, Drawing 8595404, covering a detail of Drawing 8595822.

Critical		
None defined		
Major		
101	Pitch diameter of large thread, max	Gage
102	Minor diameter of large thread, max	Gage
103	Pitch diameter of small thread, max	Gage
104	Minor diameter of small thread, max	Gage
105	Concentricity of pitch diameter of large thread with pitch diameter of small thread including perpendicularity with base of large thread cavity	Gage
106	Concentricity of hole counterbore with hole	Gage
107	Concentricity of hole with pitch diameter of small thread including perpendicularity with base of large thread cavity	Gage
108	Symmetry of flats with pitch diameter, large thread	Gage
109	Diameter of hole	Gage
110	Diameter of hole counterbore	Gage
111	Diameter of pin holes	Gage
112	Depth of large thread cavity	Gage

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Characteristic	Defect	Method of Inspection
113	Depth of small thread cavity	Gage
114	Length of effective large thread, min	Gage
115	Length of effective small thread, min	Gage
116	Thickness of wall	Gage
117	Height, max	Gage
118	Internal radius at base of large thread cavity missing or incorrect	Visual
119	Thread damaged or not full	Visual
120	Metal defective	Visual
121	Protective finish inadequate or defective	Visual
122	Surface finish of hole improper	Visual
123	Burr	Visual
124	Foreign matter present	Visual
Minor		
201	Pitch diameter of large thread, min	Gage
202	Pitch diameter of small thread, min	Gage
203	Pitch diameter of set screw thread	Gage
204	Minor diameter of set screw thread, max	Gage
205	Diameter of boss	Gage
206	Diameter of flange holes	Gage
207	Diameter of groove	Gage
208	Depth of hole counterbore	Gage
209	Total length	Gage
210	Length between flange holes	Gage
211	Length of flange	Gage
212	Width across body flats	Gage
213	Width of groove	Gage
214	Width of flange	Gage
215	Thickness of flange	Gage
216	Location of pin holes	Gage
217	Location of groove	Gage
218	Location of flange hole	Gage
219	Location of set screw holes	Gage
220	Location of initiator pin hole	Gage
221	Marking missing, incorrect, or illegible	Visual
222	Radius or chamfer missing or incorrect except as otherwise classified	Visual
223	Surface finish improper, except as other classified	Visual

4.5.2.5 Pin, Safety, Drawing 8595405, covering a detail of Drawing 8595822.

Critical		
None defined		
Major		
101	Diameter of wire	Gage
102	Profile incorrect	Visual
103	Metal defective	Visual

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Characteristic	Defect	Method of Inspection
104	Protective finish inadequate or defective	Visual
105	Burr or sharp edge	Visual
106	Foreign matter present	Visual
Minor		
201	Diameter of head	Gage
202	Total length	Gage
203	Radius missing or incorrect	Visual

4.5.2.6 Spring, Initiator, Drawing 8595406, covering a detail of Drawing 8595822.

Critical		
None defined		
Major		
101	Load at specified heights	Gage
102	Height, solid, max	Gage
103	Ends incorrect	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp edges	Visual
106	Foreign matter present	Visual
Minor		
201	Diameter outside, closed, max	Gage
202	Diameter inside, min	Gage

4.5.2.7 Pin, Firing, Drawing 8596168 covering a detail of Drawing 8595822.

Critical		
None defined		
Major		
101	Concentricity of body with cavity	Gage
102	Concentricity of body with striker	Gage
103	Diameter of body	Gage
104	Diameter of cavity	Gage
105	Diameter of striker	Gage
106	Diameter of ball hole, max	Gage
107	Depth of ball hole counterbore, max	Gage
108	Location of datum diameter	Visual
109	Radial profile of striker	Gage
110	Radial location of ball holes	Visual
111	Metal defective	Visual
112	Protective finish inadequate or defective	Visual
113	Surface finish improper	Visual
114	Burr or sharp edge	Visual
115	Foreign matter present	Visual

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Characteristic	Defect	Method of Inspection
Minor		
201	Diameter of cavity counterbore	Gage
202	Diameter of ball hole, min	Gage
203	Diameter of ball hole counterbore	Gage
204	Diameter of midsection	Gage
205	Depth of cavity counterbore	Gage
206	Depth of cavity	Gage
207	Depth of ball hole counterbore, min	Gage
208	Total length	Gage
209	Length from body to striker	Gage
210	Length to ball hole	Gage
211	Drill point cavity, max	Visual
212	Radius or chamfer missing or incorrect	Visual

4.5.2.8 Housing, Firing Pin, Drawing 8596169, covering a detail of Drawing 8595822.

Critical

None defined

Major		
101	Pitch diameter of small thread, min	Gage
102	Major diameter of small thread, min	Gage
103	Pitch diameter of large thread, min	Gage
104	Major diameter of large thread, min	Gage
105	Concentricity of counterbore with cavity	Gage
106	Concentricity of flange with pitch diameter of small thread	Gage
107	Concentricity of groove with flange	Gage
108	Concentricity of striker hole with cavity	Gage
109	Concentricity of pitch diameter of large thread with cavity	Gage
110	Concentricity of cavity with pitch diameter of small thread, including perpendicularity with face of flange	Gage
111	Perpendicularity of boss face with pitch diameter of large thread	Gage
112	Diameter of cavity, max	Gage
113	Diameter of external datum location	Gage
114	Diameter of groove	Gage
115	Diameter of flange, min	Gage
116	Diameter of small thread undercut, min	Gage
117	Length from rear flange to cavity	Gage
118	Length to internal datum diameter	Gage
119	Width of rear flange	Gage
120	Angle of external datum diameter	Gage
121	Angle at internal datum diameter	Gage
122	Height of boss	Gage
123	Profile of groove	Gage
124	Radius at root of boss	Gage
125	Thread damaged or not full	Visual
126	Metal defective	Visual

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characteristic	Defect	Method of Inspection
127	Protective finish inadequate or defective	Visual
128	Surface finish in cavity improper	Visual
129	Burr or sharp corner	Visual
130	Foreign matter present	Visual
Minor		
201	Pitch diameter of small thread, max	Gage
202	Pitch diameter of large thread, max	Gage
203	Diameter of cavity, min	Gage
204	Diameter of flange, max	Gage
205	Diameter of body	Gage
206	Diameter of counterbore	Gage
207	Diameter of undercut in cavity, if applicable	Gage
208	Diameter of striker hole	Gage
209	Diameter of small thread pilot	Gage
210	Depth to bottom of cavity	Gage
211	Total length	Gage
212	Length from small thread pilot to flange	Gage
213	Length across flanges	Gage
214	Length of body	Gage
215	Width of undercut in cavity, if applicable	Gage
216	Width of undercut, small thread	Gage
217	Width of groove	Gage
218	Location of flange	Gage
219	Radius or chamfer missing or incorrect, except as otherwise classified	Visual
220	Surface finish improper, except as otherwise classified	Visual

4.5.2.9 Seal, Drawing 10524102 covering a detail of Drawing 8595822.

Critical		
None defined		
Major		
101	Diameter outside, min	Gage
102	Width, min	Gage
103	Thickness, min	Gage
104	Cut, cracked, or split	Visual
105	Flash	Visual
106	Foreign matter present	Visual
Minor		
201	Diameter outside, max	Gage
202	Width, max	Gage
203	Thickness, max	Gage

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4.5.2.10 Assembly, Drawing 8595822, less 8595403, 8595405, 8593013, and 8593014.

Characteristic	Defect	Method of Inspection
Critical		
1	Protrusion of firing pin	Gage
2	Steel ball missing	Gage
Major		
None defined		
Minor		
None defined		

4.5.2.11 Assembly, Drawing 8595822.

Critical		
1	Any part missing, except as otherwise classified	Visual
2	Improper assembly	Visual
Major		
101	Initiator pin not free to rotate	Manual
102	Safety pin loose or deformed	Manual/Visual
103	Marking missing, incorrect, or illegible	Visual
Minor		
201	Protective cap missing	Visual
202	Protective finish inadequate or defective	Visual

4.5.2.12 Box, Fiberboard (Inner) Packing Ammunition Drawing 8796541

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Cut or damaged	Visual
203	Poor workmanship	Visual

4.5.2.13 Box, Packing, Ammunition, Drawing 8796542.

Critical		
None defined		
Major		
None defined		

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Characteristic	Defect	Method of Inspection
Minor		
201	Contents loose	Manual
202	Broken or damaged	Visual
203	Number of units contained incorrect	Visual
203	Marking missing, incorrect, or illegible	Visual

4.5.2.14 Box, Fiberboard (Outer) Alternative, Packing, Drawing 9216934.

Critical

None defined

Major

None defined

Minor

201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Hardness. Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each subassembly shall be subjected to the hydrostatic pressure test. Failure of any subassembly to comply with 3.5 shall be cause for rejection of the subassembly.

4.5.3.3 Firing pin release. The initiator pin of each subassembly shall be subjected to the firing pin release test. Failure of any subassembly to comply with 3.6 shall be cause for rejection of the subassembly.

4.5.3.4 Initiator pin security. The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any subassembly to comply with 3.7 shall be cause for rejection of the subassembly.

4.5.3.5 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.3.6 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit in the inspection sample to comply with 3.10 shall be cause for rejection of the assembly as specified in 4.6.6.

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4.6 Test methods and procedures.

4.6.1 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.6.2 Hydrostatic pressure. Prior to the application of protective finish, the cap and chamber shall be mounted in the device, using a fixture for properly sealing all openings and subjected to the internal hydrostatic pressure to determine compliance with 3.5.

4.6.3 Firing pin release. The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

4.6.4 Initiator pin security. The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.6.5 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.6 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degrees rotation of the assembly for four full passes.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING.

5.1 Packing and marking. Packing and marking shall be levels A or B as specified by the contracting activity:

Level A	In accordance with Drawings 8796541 and 8796542
Level B	In accordance with Drawings 8796541 and 9216934

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-932-5031.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)

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c. If first article inspection is required (see 3.2 and 6.3)
d. The responsible inspection activity (see 4.1)
e. Level of marking and packing (see 5.1)

6.3 **First article.** When a first article inspection is required, the items should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish, and 50 shear pins. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 **Definition.**

6.4.1 **Mechanical failure.** Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 **Subject term (keyword) listing.**

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 **Changes from previous issue.** Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy-OS
Air Force-99

Preparing activity:
Navy-OS
(Project 1377-0959)

Review activity:
Air Force-70

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